Journal of the European Ceramic Society 40 (2020) 4774–4782

Contents lists available at ScienceDirect

Journal of the European Ceramic Society

journal homepage: www.elsevier.com/locate/jeurceramsoc

Small scale fracture and strength of high-entropy carbide grains during microcantilever bending experiments

Tamás Csanádi[a,*], Marek Vojtko[a], Zoltán Dankházi[b], Michael J. Reece[c], Ján Dusza[a,d]

[a] Institute of Materials Research, Slovak Academy of Sciences, Watsonova 47, 04353 Košice, Slovak Republic
[b] Department of Materials Physics, Eötvös Loránd University, Pázmány P. sétány 1/A, 1117 Budapest, Hungary
[c] School of Engineering and Material Science, Queen Mary University of London, Mile End Road, London, E1 4NS, UK
[d] Donát Bánki Faculty of Mechanical and Safety Engineering, Óbuda University, Népszínház utca 8, 1081 Budapest, Hungary

ARTICLE INFO

Keywords:
Fracture strength
fracture surface
microcantilever bending
high-entropy carbide
fracture toughness

ABSTRACT

The fracture behaviour of (Hf-Ta-Zr-Nb)C high-entropy carbide grains was investigated by microcantilever bending experiments, and fracture related properties (e.g. strength, toughness) were determined using linear beam theory. Microcantilevers were FIB-milled from large grains with {001} and {101} orientations and were subjected to micro-bending experiments. SEM based fractographic analysis of the broken cantilevers revealed that approximately half of them fractured at the fixing position at FIB-induced surface cracks, while the rest of the beams failed at small cracks located at submicron size pores or inclusions. In all cases, fracture occurred on the {001} cleavage plane. The fracture strength of beams fractured at the fixing position was 11.8 ± 0.2 GPa, while the strength of the beams that failed at submicron defects was in the range of 3.8-8.9 GPa. The calculation of stress concentration in the vicinity of pores revealed that local stress field exceeded the value that induced cracking in 'defect free' beams.

1. Introduction

High-entropy ultra-high temperature ceramics are a novel group of materials, which are being developed to broaden the limited set of ultra-high temperature ceramics (UHTCs), comprising about 15 materials [1], and improve their mechanical properties to meet the continuously growing demands of ultra-high temperature applications [2,3]. In general, UTHCs are defined as materials with melting temperature above 3000 °C that can withstand extreme environments for applications above 1500 °C in oxidizing and aggressive environments, such as rocket propulsions, hypersonic and re-usable atmospheric re-entry vehicles. UHTCs are usually based on the refractory borides, carbides and nitrides of the group of IV and V transition metals. Due to the nature of their chemical bonds, these materials exhibit high hardness (> 20 GPa) and strength (> 500 MPa) with excellent oxidation and ablation resistance at elevated temperatures [4–6].

Based on the discovery of high-entropy alloys [7,8] and entropy-stabilized oxides [9], bulk high-entropy UHTCs have been successfully synthesized both from metal diborides by Gild el al. [2] and from metal carbides by the present authors [3]. These novel high-entropy ceramics are composed of four or more different transition metal elements of equimolar proportions and boron or carbon atoms, and form hexagonal or cubic solid solution structures, respectively. Experimental and theoretical studies on these materials suggest that high-entropy ultra-high temperature ceramics could broaden the limited set of UHTCs and their mechanical properties could go beyond those of the base borides [2] and carbides [3,10–13]. Regarding high-entropy carbides, Castle et al. [3] have recently reported that the lattice parameter mismatch of the component monocarbides is a key predictive factor for determining solid solution formation, and hence the (Hf-Ta-Zr-Nb)C composition formed a single phase solid solution more readily than the (Hf-Ta-Zr-Ti)C composition. The high resolution microstructural analysis exhibited a random arrangement of transition metal elements in (Hf-Ta-Zr-Nb)C at the atomic scale using energy dispersive X-ray spectroscopy and high angle annular dark field scanning transmission electron microscopy [14]. Micromechanical testing of (Hf-Ta-Zr-Nb)C grains revealed higher nano-hardness [3] and significantly enhanced yield and fracture strength during micropillar compression compared to the corresponding base monocarbides [12]. Additionally, the plasticity of (Hf-Ta-Zr-Nb)C grains was observed during micropillar compression due to the operation of $\{110\}\langle 1\bar{1}0\rangle$ slip systems [12]. High-temperature creep testing of (Hf-Ta-Zr-Nb)C samples exhibited approximately ten times lower creep rate than that of the corresponding monocarbides controlled by dislocation glide on the $\{111\}\langle 1\bar{1}0\rangle$ slip systems, dislocation

* Corresponding author.
E-mail address: tcsanadi@saske.sk (T. Csanádi).

https://doi.org/10.1016/j.jeurceramsoc.2020.04.023
Received 31 January 2020; Received in revised form 12 April 2020; Accepted 15 April 2020
Available online 20 April 2020
0955-2219/ © 2020 Elsevier Ltd. All rights reserved.

Table 1
Geometry of FIB-milled high-entropy carbide cantilevers and corresponding fracture related parameters derived from micro-bending experiments.

Beams	Orientation of z-axis	L-x_2 (μm)	a (μm)	b (μm)	m_a (μm)	x_1 (μm)	Location of fracture	Pore and crack sizes $2r$, l (μm)	Fracture strength σ (GPa)	Stress concentration σ_c (GPa)	Fracture toughness K_{Ic} (MPam$^{0.5}$)
#1	{001}	11.21	3.43	3.31	1.16	1.23	defect	-	8.9	-	-
#2	{001}	15.97	3.55	2.73	1.37	0	fixing	-	11.6	-	-
#3	{001}	15.91	3.29	1.91	1.40	3.18	defect	>0.31, <0.06	5.9	13.8	3.2-4.5
#4	{001}	13.73	2.74	1.84	1.19	0	fixing	-	12.0	-	-
#5	{101}	13.45	2.58	1.85	1.14	0	fixing	-	11.9	-	-
#6	{101}	14.42	3.21	1.62	1.46	1.27	defect	>0.17, <0.03	7.8	18.4	3.1-4.2
#7	{101}	13.37	3.25	1.84	1.49	4.81	defect	>0.52, <0.16	3.8	8.0-19.0	2.4-4.3

climb and the formation of voids and cracks [13].

Although high-entropy carbides exhibit improved mechanical properties compared to base monocarbides, their low ductility might be a drawback in such high hardness materials. At room temperature, group IV transition metal carbides (TMCs) (TiC, ZrC, HfC) were reported to slip on $\{110\}\langle 1\bar{1}0\rangle$ and be more brittle than group V TMCs (VC, NbC, TaC) which slip on $\{111\}\langle 1\bar{1}0\rangle$ due to the metallic like character of their bonds [15,16]. Therefore, the deformability of high-entropy carbides is strongly limited by the operating slip systems and initial defect structure of the grains, similar to other refractory materials like WC-Co [15–18]. Microstructural flaws, such as pores and inclusions, and other obstacles in the material (e.g. grain boundaries) that hinder the movement of dislocation and limit plastic deformation can act as stress raisers under tensile stresses, initiating cracks and leading to fracture [19]. Based on ab initio simulations, it was predicted that cleavage is preferred on {001} planes in stoichiometric transition metal carbides [20]. In order to develop novel high-entropy carbides with improved fracture resistance, it is important to understand the fracture behaviour of their grains and grain boundaries using direct micromechanical investigations. This has been successfully carried out using microcantilver bending experiments on brittle materials in the recent years [21–25]. To the best of our knowledge, the fracture behaviour of high-entropy carbide grains has not been studied by microcantilever bending up to now. Thus, the aim of the present work is to study the fracture behaviour of high-entropy carbide grains using microcantilever bending experiments and to determine fracture related properties such as strength and toughness.

2. Experimental

Bulk, single phase (Hf-Ta-Zr-Nb)C high-entropy carbide sample was synthesized from high purity raw powders (HfC and TaC from H.C. Starck, ZrC and NbC from American Elements) using ball milling and spark plasma sintering (SPS). The carbide powders were weighed in equiatomic proportions and mixed at 200 rpm for 24 hr in WC pots. Then the powder mixture was compacted into a 20 mm diameter graphite die and SPS processed (FCT HPD 25, FCT Systeme GmbH) using a two-step optimised heating profile with a 10 min dwell at 1800 °C and a 2 min dwell at 2300 °C in a vacuum of ~5 Pa. The loading pressure was increased from the minimum of 16 MPa to 40 MPa during the dwell at 1800 °C, and was reduced back down to 16 MPa at 2300 °C. For more details the reader is referred to our recent paper [3]. The sample was subjected to standard metallographic procedures, including cutting, grinding and polishing on a Struers machine. The final polishing was carried out on a velvet polishing disc using a polycrystalline diamond suspension (crystal size of 0.1 μm).

The crystal orientation of (Hf-Ta-Zr-Nb)C grains was determined from the EBSD maps using the Orientation Imaging (OIM) software. The program describes the orientation of the crystals based on Euler angles (φ_1,Φ,φ_2) relative to the sample coordinate system. Further details on the sample preparation and EBSD analysis can be found in our earlier work [12].

Microcantilevers were fabricated using a focused ion beam (FIB) technique in a dual-beam FIB/SEM Zeiss Auriga microscope operated at 30 kV. Milling was carried out in three consecutive steps, which included the preparation of trenches, cutting and final polishing of the micocantilevers with the corresponding currents of 20 nA, 2 nA and 500 pA, respectively. Ten microcantilevers were fabricated from large grains of {001} and {101} orientations based on EBSD maps. The microcantilevers had pentagonal cross-sections with their geometrical parameters given in Table 1, which were measured by SEM as shown schematically in Fig. 1.

The microcantilever bending tests were carried out on an Agilent G200 NanoIndenter equipped with a diamond spherical tip with a nominal radius of 1 micron, using a load rate of 0.025 mN/s. The maximum load was set to 10 mN, which was sufficiently high to induce fracture. All of the tests were performed until fracture occurred, which was recognized on the load-displacement data as a sudden increment in displacement together with a drop in the load. The data recorded were corrected for drift and a slight displacement caused by an indentation effect, similar to that reported in [18]. The microcantilevers were subjected to SEM (Zeiss Auriga) before and after testing to measure the beam geometry and to analyze the fracture surfaces, respectively. For

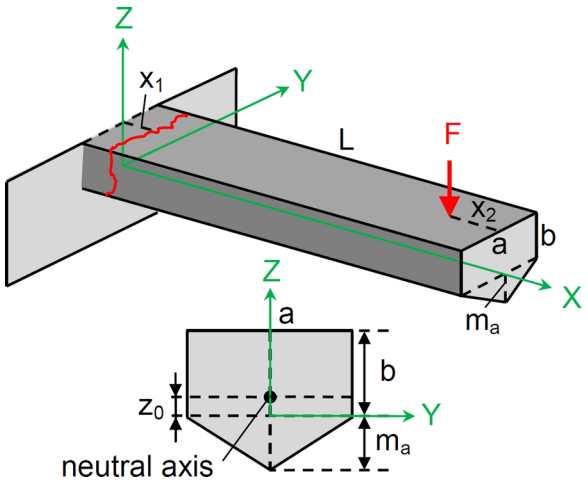


Fig. 1. Schematic of a microcantilever sample, indicating the geometrical parameters (L, a, b, m_a, z_0) and location of fracture (x_1) caused by an applied load of F at a distance of x_2 from the free end (reproduced with copyright permission from ref. [18] by Elsevier).